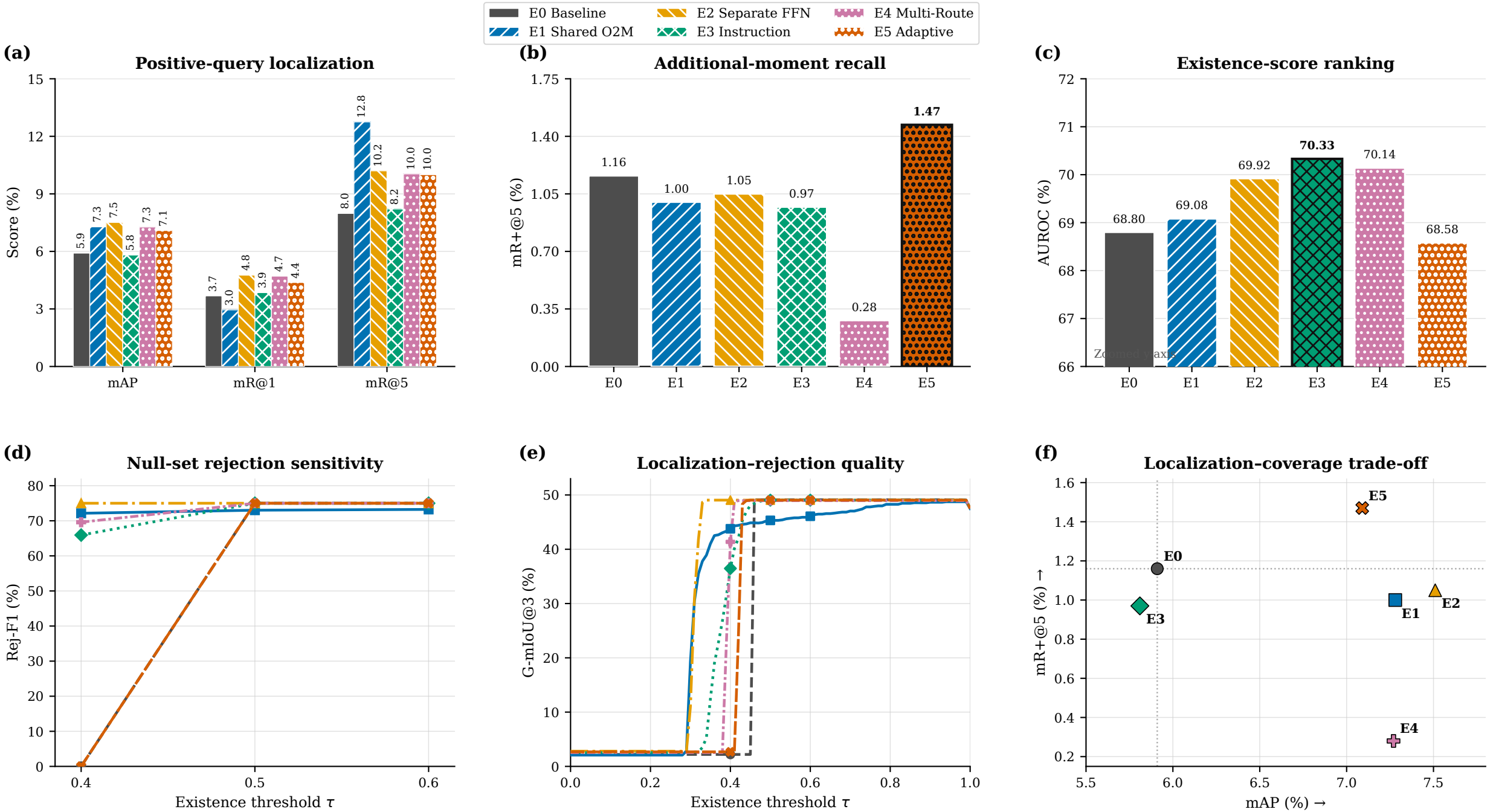

E0 Baseline
E1 Shared O2M
E2 Separate FFN
E3 Instruction
E4 Multi-Route
E5 Adaptive
(a)
Positive-query localization
Score (%)
mAP
mR@1
mR@5
5.9
7.3
7.5
5.8
7.3
7.1
3.7
3.0
4.8
3.9
4.7
4.4
8.0
12.8
10.2
8.2
10.0
10.0
(b)
Additional-moment recall
mR+@5 (%)
1.16
1.00
1.05
0.97
0.28
1.47
E0
E1
E2
E3
E4
E5
(c)
Existence-score ranking
AUROC (%)
Zoomed y-axis
68.80
69.08
69.92
70.33
70.14
68.58
(d)
Null-set rejection sensitivity
Rej-F1 (%)
Existence threshold τ
(e)
Localization-rejection quality
G-mIoU@3 (%)
Existence threshold τ
(f)
Localization-coverage trade-off
mR+@5 (%) →
mAP (%) →
E0
E1
E2
E3
E4
E5